The notes cover the following points regarding **Data Visualization** and related data science processes:

- **Dep**: Tuesday
- **Topic**: #Data Visualization
- **Key Points**:
  - How to integrate **pyexplainer** to perform **data cleaning model building** and due to algorithms to perform **explainer** and **explainer** ML Models
- **Details**: Organize/edit/add/ref/ed to ?

You would like me to elaborate on any of these **data science** concepts, such as **feature engineering** or the ML process?

Ask anything, create images, or analyze files

AI responses may include mistakes.

this will wants to integrate in power bi tools to use algorithm and code

2 sites

Yes, you can integrate PyCaret into Power BI to utilize its data cleaning, model building, and creating custom visualizations. This functionality bridges the gap between Python's powerful data science libraries and Power BI's reporting capabilities.

How to Integrate PyCaret in Power BI

The process involves two key steps:

- **Prerequisites:** You need both Power BI Desktop and a compatible version of Python installed on your machine. You also need essential libraries like `pandas`, `flask` (for manipulation), and `matplotlib` or `seaborn` (for visualizations) installed via `pip`.
- **Enable Scripting:** In Power BI Desktop, you must enable Python scripting in the options. Go to **File > Options and settings > Options > Python scripting** and specify the installation path of your Python environment.

Use Python for Data Tasks:

- **Data Import/Transformation:** You can use Power BI's "Get Data" feature to load data. Get data > "More..." > "Other" > run scripts within the Power Query Editor (`Transform > Run Python Script`) to perform advanced data cleaning and transformations. The data must be in a pandas DataFrame to be used in Power BI.
- **Model Building/Explainer:** Run the code for calculating the PI scores, you can run Python scripts in Power BI using the `exec(open` to build and evaluate machine learning models, then export the results to Power BI as a DataFrame for visualizations.

Custom Visuals: You can add a Python visual to your report canvas and use it to configure the **Power BI** environment for **Python** scripting?

this method can see generate charts and graphs in power bi

Yes, you can **generate** charts and graphs in Power BI using the **Python/matplotlib** method. This allows you to leverage the extensive visualization libraries of Python.

Generating Charts and Graphs

- **Python Visual Element:** Power BI provides a specific Python visual element in the visualization pane that you can drag onto your report canvas.
- **Script Editor:** Once placed, an editor window appears where you can enter Python code. Power BI automatically passes the necessary data from your report into a `pandas` DataFrame called `dataset`.
- **Libraries:** You can standard Python visualization libraries within this script:
  - **Matplotlib:** A foundational library with a wide range of charts.
  - **Seaborn:** Built on Matplotlib, offering more advanced statistical plots.
  - **Plotly/Dash:** Some interactive libraries might require specific configurations or may only work in Power BI Desktop and not the online service.
- **Output:** Your Python script generates the chart, which Power BI then interprets and Power BI renders the final image and renders it on the report canvas.

Limitations

- **Static Images:** The generated visuals are typically static images. They usually do not interact dynamically with Power BI's slicers on the rest of the Power BI report. Interactivity is limited to the visuals itself. The underlying data for the Python visual will update when filters are applied, but the visual itself remains a non-interactive image.

Would you like me to see an example of a simple Python script used to generate a **chart** in Power BI?

how to perform cleaning and plotting in python scripting on ...

Performing **data cleaning** and **plotting** with Python in Power BI involves using two separate areas: the Power Query Editor for cleaning and the report canvas for plotting. [1]

Data Cleaning with Python in Power Query

You can Python scripts within the Power Query Editor to clean your data before it is loaded into the main data model. The `pandas` library is essential here. [2]

- **Access the Editor:** In Power BI Desktop, click **Transform data** in the ribbon.
- **Run Script:** In the Power Query Editor, select **Transform > Run Python Script**.
- **Scripting:** Your data is automatically provided to the script as a pandas DataFrame named `dataset`. You can then apply various data cleaning functions. [2]

```
# Get data
# values in 'Sales' column with
df['Sales'].fillna(0)

# Remove rows with missing
df.dropna()

# Convert text to lowercase
df['Category'].str.lower()
df['Customer Name'].str.
# Remove leading/trailing spaces
df['Product Name'].str.strip()
```

Use code with caution.

Plotting with Python Visuals in Power BI

After cleaning the data, you use the Python visual elements in Power BI's report canvas to create charts. You will need `matplotlib` and `seaborn` installed in your Python environment. [3]

- **Add Visual:** On the report canvas, select the **Python visual** icon from the Visualizations pane.
- **Add Data Fields:** Drag the data fields you want to plot into the "Values" section. This populates the `dataset` with the data. [4]
- **Scripting:** Enter your plotting code in the Python script editor that appears at the bottom of the screen. [3]

```
import matplotlib.pyplot as plt
import seaborn as sns

# Create a bar chart
plt.figure(figsize=(8, 5))
sns.barplot(data=dataset, x='
plt.title('Sales by Category')
plt.show()
```

Use code with caution.

Would you like to know more about the **specific Python libraries** needed for these operations?

can this same method to generate plotting and graphs

Yes, absolutely, that same method is precisely how you generate plots and graphs using Python within Power BI.

You use the Python **visual** element on the report canvas to create charts and graphs. The process is the same as using Python in Power Query: you write a script, and Power BI automatically provides the data you need to visualize. The data provided to the visual. Power BI then renders the output as a static image chart on your dashboard.

Would you like to explore some specific examples of different types of **charts** you could create, like a histogram or a line plot?

Give some examples for performs cleaning operations and data ...

Certainly, here are examples of common Python scripts used for **data cleaning** in Power Query and **data visualization** in the Power BI report canvas.

Data Cleaning Examples (Power Query Editor)

These scripts are run in the **Run Python Script** window within the Power Query Editor. Remember that the input data is automatically loaded into a pandas DataFrame called `dataset`.

Example 1: Handling Missing Values

This script fills numerical values (`fillna`) in the `Quantity` column with the median of that column and missing values in the `Category` column with the string `'Unknown'`.

```
import pandas as pd
import numpy as np

# Fill missing numerical values with
median_quantity = dataset['Quantity'
dataset['Quantity'] = dataset['Qua

# Fill missing categorical values
dataset['Category'] = dataset['Cat

# The resulting 'dataset' DataFra
```

Use code with caution.

Example 2: Filtering Data and Creating New Columns

This script filters out rows where `Sales` are less than 1000 and calculates a new `Profit_Margin` column.

```
import pandas as pd

# Filter rows based on a condition
dataset = dataset[dataset['Sales'

# Create a new calculated column
# Check to prevent division by zero
dataset['Profit_Margin'] = dataset

# The filtered and modified 'data
```

Use code with caution.

Data Visualization Examples (Report Canvas)

These scripts are run in the Python script editor when you drag a **Python visual** onto your report canvas. Ensure you drag the relevant fields into the **Values** section of the visual's properties first.

Example 1: Bar Chart using Seaborn

This script creates a bar chart showing the total sales for each Region.

```
import matplotlib.pyplot as plt
import seaborn as sns

# Create a bar chart
plt.figure(figsize=(8, 5))
sns.barplot(data=dataset, x='Sales'
plt.title('Total Sales by Region')
plt.show()
```

Use code with caution.

Example 2: Scatter Plot using Matplotlib

This script generates a scatter plot to visualize the relationship between `Sales` and `Profit`.

```
import matplotlib.pyplot as plt

# Create a scatter plot
plt.figure(figsize=(8, 5))
plt.scatter(dataset['Sales'], y=
plt.title('Sales vs Profit')
plt.show()
```

Use code with caution.

Would you like some examples showing how to use `sklearn` libraries for **model evaluation** metrics like the **F1 score** in Power BI?

AI responses may include mistakes.